

Simple and Multiple Linear Regression

A. Simple linear regression with R

This chapter is a brief introduction to simple and multiple linear regression and how to use this method in a real context. We present the relevant R commands and use a real data set as a connecting thread as we present the key concepts for this method. We treat the case of qualitative explanatory variables, as well as interaction of explanatory variables.

We discuss model validation with a study of residuals and mention the issue of collinearity. We also present a few methods for variable selection.

We return to the data set Birth-weight. We wish to explain the variability of child weight at birth as a function of characteristics of the mother, of family history and of behaviour during pregnancy. The explained variable is weight at birth (quantitative variable BWT, expressed in grammes); the explanatory:

This study focused on risks associated with low weight at birth; the data were collected at the Baystate Medical Centre, Massachusetts, in 1986. Physicians have been interested in low weight at birth for several years, because underweight babies have high rates of infant mortality and infant anomalies. The behaviour of the mother-to-be during pregnancy (diet, smoking habits) can have a significant impact on the chances of having a full-term pregnancy, and thus of giving birth to a child of normal weight. The data file includes information on 189 women (identification number: ID) who came to the centre for consultation. Weight at birth is categorized as low if the child weighs less than 2,500 g.

Description	Unit or coding	Variable
Age of mother	Years	AGE
Weight of mother at last menstrual period	Pounds	LWT
Race of mother	1 = white; 2 = black; 3 = other	RACE
Smoking during pregnancy	Yes = 1; no = 0	SMOKE
Number of premature births in medical history	0 = none; 1 = one; 2 = two; etc.	PTL
Medical history of hypertension	Yes = 1; no = 0	HT
Uterine irritability	Yes = 1; no = 0	UI
Number of medical consultations during first trimester	0 = none; 1 = one; etc.	FVT
Weight at birth	Grams	BWT
Weight at birth less than 2,500 g	Yes = 1; No = 0	LOW

Loading the data:

```
> mydata <- read.csv("wb.csv",header=TRUE,sep = "\t")
> summary(mydata)
```

ID	AGE	LWT
Min. : 4.0	Min. :14.00	Min. : 80.0
1st Qu.: 68.0	1st Qu.:19.00	1st Qu.:110.0
Median :123.0	Median :23.00	Median :121.0
Mean :121.1	Mean :23.24	Mean :129.8
3rd Qu.:176.0	3rd Qu.:26.00	3rd Qu.:140.0
Max. :226.0	Max. :45.00	Max. :250.0

RACE	SMOKE	PTL
Min. :1.000	Min. :0.0000	Min. :0.0000
1st Qu.:1.000	1st Qu.:0.0000	1st Qu.:0.0000

Median :1.000	Median :0.0000	Median :0.0000
Mean :1.847	Mean :0.3915	Mean :0.1958
3rd Qu.:3.000	3rd Qu.:1.0000	3rd Qu.:0.0000
Max. :3.000	Max. :1.0000	Max. :3.0000
HT	UI	FVT
Min. :0.00000	Min. :0.0000	Min. :0.0000
1st Qu.:0.00000	1st Qu.:0.0000	1st Qu.:0.0000
Median :0.00000	Median :0.0000	Median :0.0000
Mean :0.06349	Mean :0.1481	Mean :0.7937
3rd Qu.:0.00000	3rd Qu.:0.0000	3rd Qu.:1.0000
Max. :1.00000	Max. :1.0000	Max. :6.0000
BWT	LOW	
Min. : 709	Min. :0.0000	
1st Qu.:2414	1st Qu.:0.0000	
Median :2977	Median :0.0000	
Mean :2945	Mean :0.3122	
3rd Qu.:3475	3rd Qu.:1.0000	
Max. :4990	Max. :1.0000	

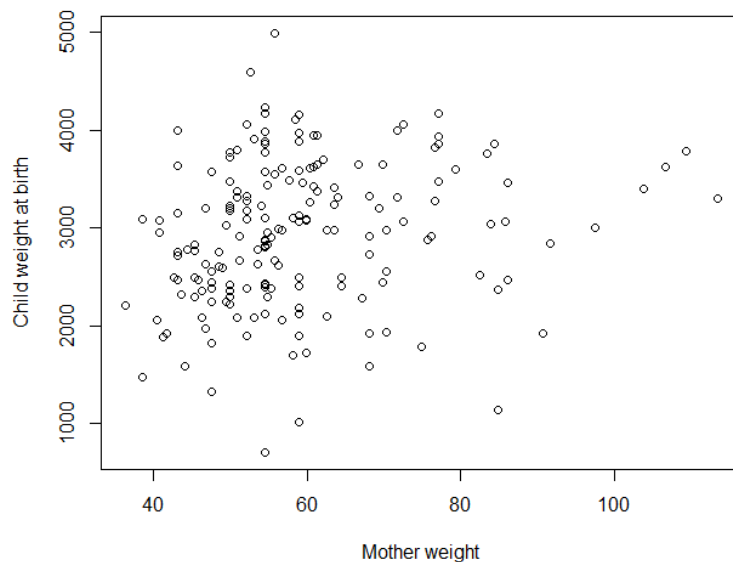
The weight of the mother is expressed in pounds. We first transform the data.frame to recode this variable in kilogrammes (1 pound = 0.45359237kg).

```
> mydata <- transform(mydata,LWT=LWT*0.4535923)
> attach(mydata)
```

1. Graphical Inspection

To study the relationship between child weight at birth and weight of the mother, we first draw the scatterplot of the points (child weight; mother weight) using the instruction plot(BWT~LWT)

```
> plot(BWT~LWT,xlab="Mother weight",ylab="Child weight at birth")
```



We observe a slight increase in child weight when mother weight increases, although this relationship is not very clear.

2. Parameter Estimation

We now study the following model:

$$BWT = \beta_1 + \beta_2 LWT + \varepsilon$$

```
> model1<-lm(BWT~LWT)
> model1
```

```
Call:
lm(formula = BWT ~ LWT)
```

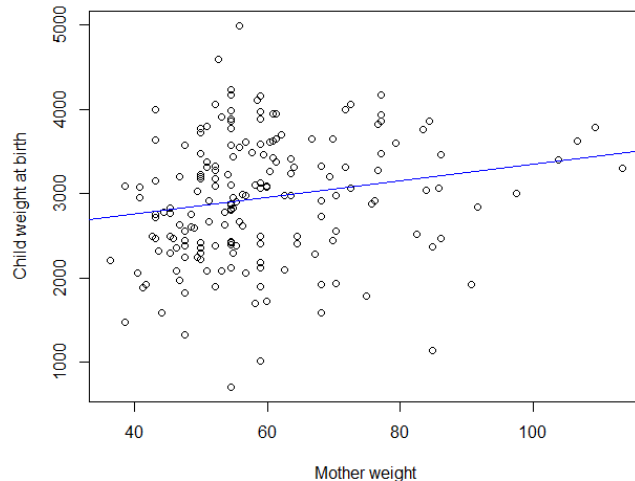
Coefficients:

(Intercept)	LWT
2369.672	9.765

The above R output gives the least squares estimates of $\hat{\beta}_1 = 2369.672$ and $\hat{\beta}_2 = 9.765$

We can now draw the regression line on the scatter plot, using the function *abline()*

```
> plot(BWT~LWT,xlab="Mother weight",ylab="Child weight")
> abline(model1,col="blue")
```



3. Tests on Parameters

Note that the function *lm()* performs a complete analysis of the linear model and that you can get a summary of the calculations related to the data set with the function *summary()*.

```
> summary(model1)
```

```
Call:
lm(formula = BWT ~ LWT)
```

```

Residuals:
    Min       1Q   Median       3Q      Max
-2192.18  -503.63    -3.91   508.25  2075.53

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 2369.672    228.431  10.374  <2e-16 ***
LWT          9.765      3.777   2.586   0.0105 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 718.2 on 187 degrees of freedom
Multiple R-squared:  0.03452, Adjusted R-squared:  0.02935
F-statistic: 6.686 on 1 and 187 DF,  p-value: 0.01048

```

Here is a description of the information in this output.

Call: formula used in the model.

Residuals: descriptive analysis of residuals $\hat{\varepsilon}_i = \hat{y}_i - y_i$, we shall see that residuals are used to validate the assumptions of the regression model.

Coefficients: this table has four columns:

- Estimate gives the estimates of the parameters of the regression line.
- Std. Error gives the estimate of the standard deviation of the estimators of the regression line.
- t value gives the realization of Student's test statistic associated with the hypotheses $H_0 : \beta_i = 0; H_1 : \beta_i \neq 0$
- $\text{Pr}(>|t|)$ gives the p-value of Student's test.

Signif. codes: codes for significance levels.

Residual standard error: an estimate of the standard deviation of the noise σ and the associated degree of freedom $n - 2$.

Multiple R-squared: coefficient of determination r^2 (percentage of variation explained by the regression).

Adjusted R-squared: adjusted r_a^2 (of limited interest for simple linear regression).

F-statistic: realization of Fisher's test statistic associated with the hypotheses $H_0 : \beta_2 = 0; H_1 : \beta_2 \neq 0$. The associated degrees of freedom (1 and $n = 2$) are given, as is the p-value.

To get an estimate by confidence interval of the regression coefficients, we can use the function `confint()`.

```
> confint(model1)
```

	2.5 %	97.5 %
(Intercept)	1919.039836	2820.30429
LWT	2.314692	17.21502

4. Confidence and Prediction Intervals for a New Value

Consider a new observation x_0 of variable X for which we have not observed the corresponding value y_0 of the response variable Y . This value y_0 is unknown, since it is not observed, and is a realization of the random variable $Y_0 = \beta_1 + \beta_2 X_0 + \varepsilon_0$.

The predictor of Y_0 for the new value x_0 is given by $\hat{Y}_0^p = \hat{\beta}_1 + \hat{\beta}_2 x_0$.

We can also propose a prediction interval at level $1-\alpha$ for Y_0 , by finding two random bounds such that the random variable falls in the interval with probability $1-\alpha$:

$$\left[\hat{Y}_0^p \pm t_{1-\alpha/2}^{(n-2)} \hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}} \right]$$

Note that the realization $\hat{y}_0^p = \hat{\beta}_1 + \hat{\beta}_2 x_0$ is called the prediction of the unobserved value $y_0 = \beta_1 + \beta_2 x_0 + \varepsilon_0$.

Similarly, note that an estimator of the fixed and unknown value $E(Y_0 | X = x_0) = \beta_1 + \beta_2 x_0$ is given by $\hat{E}(Y_0 | X = x_0) = \hat{Y}_0 = \hat{\beta}_1 + \hat{\beta}_2 x_0$.

We can also propose a prediction interval at level $1-\alpha$ for $E(Y_0 | X = x_0)$,

$$\left[\hat{Y}_0 \pm t_{1-\alpha/2}^{(n-2)} \hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}} \right]$$

The function to define the prevision interval and confidence interval for a new value x_0 is *predict()*.

Use data Weight – Birth, we calculate the prediction of the weight of a baby whose mother weighs lwt = 56 kg.

```
> lwt0 <- 56
> predict(model1, data.frame(LWT=lwt0), interval="prediction")
      fit      lwr      upr
1 2916.504 1495.699 4337.309
```

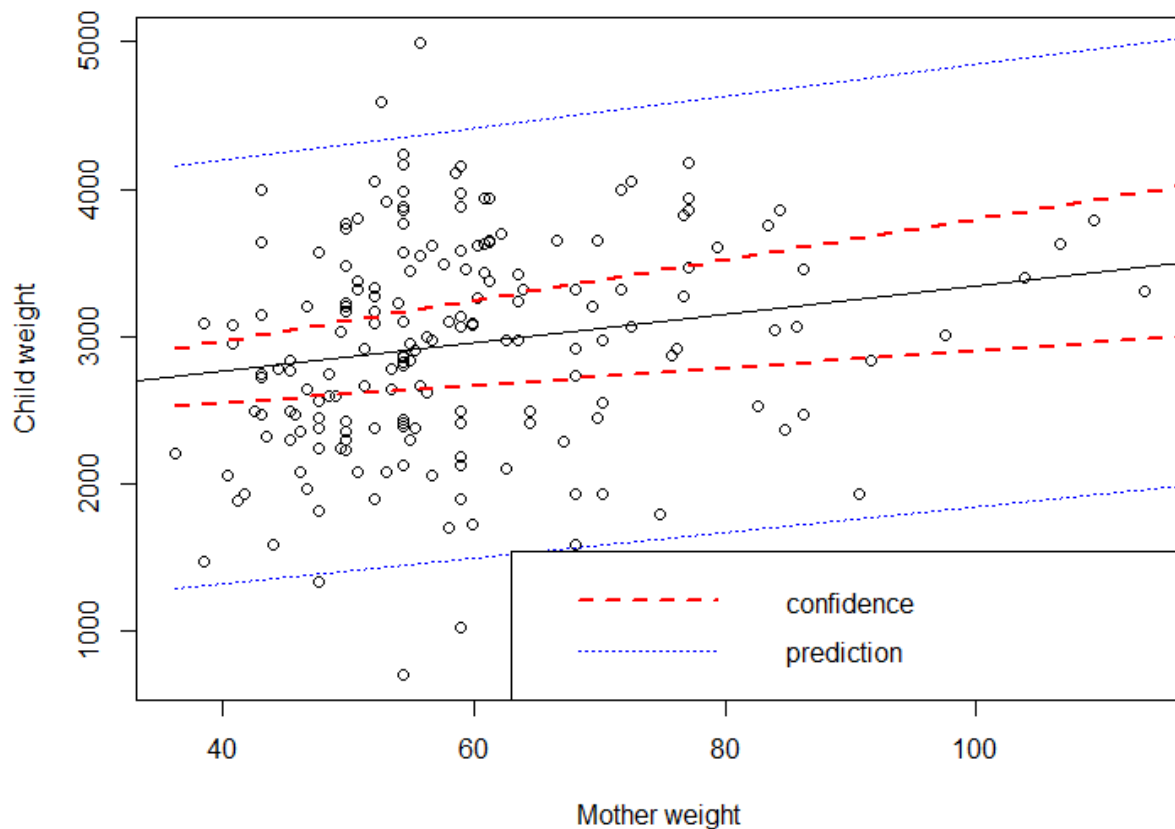
For the confidence interval of the mean value of the weight of babies with a mother weighing 56 kg:

```
> predict(model1, data.frame(LWT=lwt0), interval="confidence")
```

	fit	lwr	upr
1	2916.504	2811.225	3021.783

We now represent the confidence interval and prediction interval for a series of new values of the mother's weight

```
> x <- seq(min(LWT),max(BWT),length=50)
> predint <- predict(model1,data.frame(LWT=x),interval=
+ "prediction")[,c("lwr","upr")]
> confint <- predict(model1,data.frame(LWT=x),interval=
+ "confidence")[,c("lwr","upr")]
> plot(BWT~LWT,xlab="Mother weight",ylab="Child weight")
> abline(model1)
> matlines(x,cbind(confint,predint),lty=c(2,2,3,3),
+ col=c("red","red","blue","blue"),lwd=c(2,2,1,1))
> legend("bottomright",lty=c(2,3),lwd=c(2,1),
+ c("confidence","prediction"),col=c("red","blue"))
```



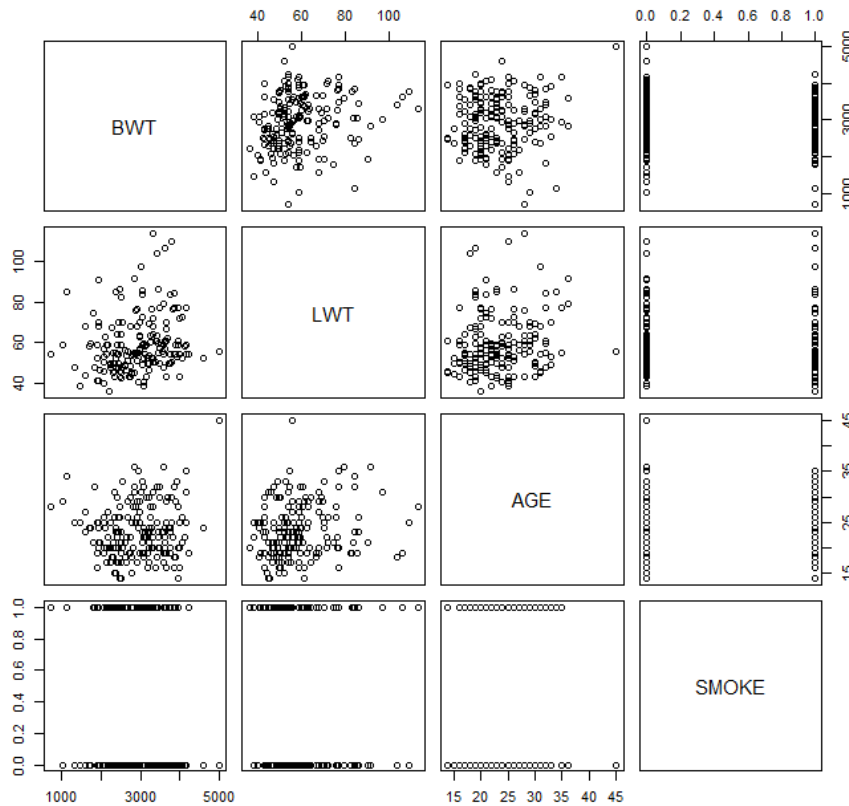
B. Multiple Linear Regression

1. Graphical Inspection

With data in part A, we make regression of child weight at birth as a function of mother age, weight and smoking status during pregnancy.

Before estimating the model, we present a scatter plot of all pairs of variables:

```
> pairs(BWT~LWT+AGE+SMOKE)
```



2. Parameter Estimation

As for simple linear regression, the model is estimated using function *lm()*:

```
> model12 <- lm(BWT~AGE+LWT+SMOKE)
> model12
```

Call:

```
lm(formula = BWT ~ AGE + LWT + SMOKE)
```

Coefficients:

(Intercept)	AGE	LWT	SMOKE
2362.720	7.093	8.860	-267.213

3. Tests on Parameters

Tests on parameters are performed by function *summary()*.

```
> summary(model12)
```

```

Call:
lm(formula = BWT ~ AGE + LWT + SMOKE)

Residuals:
    Min       1Q   Median       3Q      Max
-2069.89  -433.18    13.67   516.45  1813.75

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 2362.720    300.687   7.858 3.11e-13 ***
AGE           7.093      9.925   0.715  0.4757
LWT          8.860      3.791   2.337  0.0205 *
SMOKE       -267.213    105.802  -2.526  0.0124 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 708.8 on 185 degrees of freedom
Multiple R-squared:  0.06988, Adjusted R-squared:  0.05479
F-statistic: 4.633 on 3 and 185 DF, p-value: 0.003781

```

The results output by `summary()` are presented in the same fashion as for simple linear regression. Parameter estimates are given in the column **Estimate**.

The realizations of Student's test statistics associated with the hypotheses $H_0 : \beta_2 = 0; H_1 : \beta_2 \neq 0$ are given in column **t-value**; the associated p-values are in column **Pr(>|t|)**. Residual standard error gives the estimate of and the number of associated degrees of freedom $n - p - 1$. The coefficient of determination R^2 (**multiple R-squared**) and an adjusted version (**adjusted R-squared**) are given, as are the realization of Fisher's global test statistic (**F-statistic**) and the associated p-value.

3. Interpreting Results from Study "Weight at Birth"

Given the result of Fisher's global test (p-value D 0.003781), we can conclude that at least one of the explanatory variables is associated with child weight at birth, after adjusting for the other variables. The individual Student tests indicate that:

- Mother weight is linearly associated with child weight, after adjusting for age and smoking status, with risk of error less than 5 % (p-value = 0.0205). At same age and smoking status, an increase of 1 kg in mother weight corresponds to an increase of 8.860 g of average child weight at birth.
- The age of the mother is not significantly linearly associated with child weight at birth when mother weight and smoking status are already taken into account (p-value = 0.20661).
- Weight at birth is significantly lower for a child born to a mother who smokes, compared to children born to non-smoker mothers of same age and weight, with a risk of error less than 5 % (p-value = 0.012). At same age and mother weight, child weight at birth is 267.213 g less for a smoker mother than for a non-smoker mother.

4. Interpreting Results from Study “Weight at Birth”

Suppose we wish to predict the weight at birth of a child whose mother is 23 years old, weighs 57 kg and smokes. The function `predict()` gives a prediction, a prediction interval and a confidence interval for the mean weight of children whose mothers have these characteristics.

```
> newdata <- data.frame(AGE=23,LWT=57,SMOKE=1)
> predict(model2,newdata,interval="pred")
      fit      lwr      upr
1 2763.693 1355.943 4171.444
> predict(model2,newdata,interval="conf")
      fit      lwr      upr
1 2763.693 2600.914 2926.472
```

5. Testing a Linear Sub-hypothesis: Partial Fisher Test

Fisher’s partial test is used to test the contribution of a subset of explanatory variables in a model which already includes other explanatory variables. For example, consider the following two models:

Model 1: $BWT = \beta_1 + \beta_2 LWT + \varepsilon$

Model 2: $BWT = \beta_1 + \beta_2 LWT + \beta_3 AGE + \beta_4 SMOKE + \varepsilon$

Fisher’s test is used to test the joint contribution of variables *AGE* and *SMOKE* in model 2. The hypotheses of the test are $H_0: \beta_3 = \beta_4 = 0$ and H_1 : at least one of the coefficients β_3 or β_4 is non-zero. The following instructions are used for this test:

```
> anova(model1,model2)
Analysis of Variance Table
```

```
Model 1: BWT ~ LWT
Model 2: BWT ~ AGE + LWT + SMOKE
  Res.Df    RSS Df Sum of Sq    F Pr(>F)
1     187 96468171
2     185 92935223   2    3532949 3.5164 0.03171 *
```

```
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

The p-value of the test ($\text{Pr}(>F)=0.03171$) indicates that at least one of the two variables *AGE* or *SMOKE* gives extra information to predict child weight at birth, when mother weight has already been taken into account.

Exercises

1) This question involves the use of simple linear regression on the Auto data set.

(a) Use the `lm()` function to perform a simple linear regression with *mpg* as the response and *horsepower* as the predictor. Use the `summary()` function to print the results. Comment on the output.

For example:

i. Is there a relationship between the predictor and the response?

- ii. How strong is the relationship between the predictor and the response?
- iii. Is the relationship between the predictor and the response positive or negative?
- iv. What is the predicted mpg associated with a horsepower of 98? What are the associated 95 % confidence and prediction intervals?

(b) Plot the response and the predictor. Use the `abline()` function to display the least squares regression line.

(c) Use the `plot()` function to produce diagnostic plots of the least squares regression fit. Comment on any problems you see with the fit.

2) In this exercise you will create some simulated data and will fit simple linear regression models to it. Make sure to use `set.seed(1)` prior to starting part (a) to ensure consistent results.

(a) Using the `rnorm()` function, create a vector, `x`, containing 100 observations drawn from a $N(0; 1)$ distribution. This represents a feature, X .

(b) Using the `rnorm()` function, create a vector, `eps`, containing 100 observations drawn from a $N(0; 0.25)$ distribution|a normal distribution with mean zero and variance 0.25.

(c) Using `x` and `eps`, generate a vector `y` according to the model $Y = -1 + 0.5X + \epsilon$: (3.39)

What is the length of the vector `y`? What are the values of β_0 and β_1 in this linear model?

(d) Create a scatterplot displaying the relationship between `x` and `y`. Comment on what you observe.

(e) Fit a least squares linear model to predict `y` using `x`. Comment on the model obtained. How do $\hat{\beta}_0$ and $\hat{\beta}_1$ compare to β_0 and β_1 ?

(f) Display the least squares line on the scatterplot obtained in (d).

Draw the population regression line on the plot, in a different color. Use the `legend()` command to create an appropriate legend.

(g) Now fit a polynomial regression model that predicts `y` using `x` and `x^2`. Is there evidence that the quadratic term improves the model fit? Explain your answer.

(h) Repeat (a){(f) after modifying the data generation process in such a way that there is less noise in the data. The model (3.39) should remain the same. You can do this by decreasing the variance of the normal distribution used to generate the error term in (b). Describe your results.

(i) Repeat (a)-(f) after modifying the data generation process in such a way that there is more noise in the data. The model (3.39) should remain the same. You can do this by increasing the variance of the normal distribution used to generate the error term in (b). Describe your results.

(j) What are the confidence intervals for β_0 and β_1 based on the original data set, the noisier data set, and the less noisy data set? Comment on your results.